



DC – AI – Routing – Switching – Security – Wireless – Automation

MX301 Deepdive

An HPE Juniper Networking Techpost

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HPE



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1 MX301 Deepdive

David Roy - 11/24/2025

Let's explore the capabilities of the Juniper Networks MX301 Universal Routing Platform, a 1RU edge router built on Trio 6 silicon that delivers up to 1.6 Tbps full-duplex throughput, supports a broad range of interface speeds from 1GE to 400GE, and integrates features like hardware-accelerated MACsec/IPsec. The article details system architecture, chassis design, port mapping, and targeted use-cases—highlighting how the MX301 extends the MX10K family into more compact deployments for both enterprise and service-provider environments.

1.1 Introduction

The Juniper Networks MX301 Universal Routing Platform is a compact, high-performance 1RU edge router designed for next-generation networks. It's the newest member of the MX family. Delivering up to 1.6 Tbps full-duplex throughput with exceptional density and efficiency, it consumes as little as 0.3 W per Gbps, making it ideal for service providers, cloud operators, MSOs, and enterprises at scale.

Built on Trio 6 silicon, the MX301 supports a wide range of high-density interfaces (1GE to 400GE) for flexible deployment at the WAN edge and AI edge clustering. Integrated hardware acceleration provides uncompromised security with line-rate MACsec on all ports and inline IPsec encryption. At the same time, Junos OS ensures seamless compatibility with existing MX Series platforms and full lifecycle automation. MX301 is also Class C Timing compliant.

The MX301 will launch in Jan-2026 with Junos 25.4R1 with the same feature set as MX304. The recommended release for the MX301 will be 25.4R1-S2.

1.2 Targeted use-cases

The MX301 is designed to address a wide range of deployment scenarios, providing flexibility and high performance in a compact 1RU form factor. Its targeted use cases can be grouped into two main segments:

1.2.1 Enterprise

The MX301 supports critical enterprise network functions, including:

- Enterprise WAN Edge
- Enterprise Internet Gateway (Peering)
- Enterprise CPE
- Enterprise Backbone (Core)
- Data Center Gateway (DC-GW) and DC Edge

1.2.2 Service Provider

For service providers, the MX301 is well-suited for a variety of roles, such as:

- Small and Medium Peering Routers

- Distributed BNG (Broadband Network Gateway)
- High-speed Edge
- Internet PE
- Business PE
- Route Reflector

With this broad range of supported use cases, the MX301 delivers the performance, versatility, and reliability required for both enterprise and service provider environments. Let’s take a closer look at the MX301.

1.3 Chassis design

The MX301 chassis model is MX301-HW-BASE. It has a 1RU form factor, and its front panel is shown in the figure below.



Figure 1: Introducing the MX301

The key chassis specifications are listed in Table 1 below.

Table 1: MX301 Chassis Specifications*

Specification	Details
Dimensions (W x H x D)	1.75 x 17.3 x 17.7 inches (4.45 x 44 x 45 cm)
Maximum weight	66.1 lbs (30 kg)
Rack mounting	2/4-post rack mounting
Power system rating	2 PSMs, AC, DC, HV-AC/DC (1+1 redundancy)
Typical power consumption	<300 W Typical (275W to 300W at 25C, not including optics power)
Operating temperatures	32deg to 115deg F (0deg to 46deg C) at sea level

The MX301-HW-BASE is a fixed form factor, so only a few components are FRUs, as listed in the following table:

Table 2: List of MX301's FRUs

FRU Component	SKU	Installation / Slot	Hot-Swappable
Fan Tray	JNP-FAN3-1RU-BB	Rear, horizontally mounted	Yes
Power Supply	JPSU-850WAC/DC/HW-AFO-BB	Rear PSU slots (0, 1)	Yes

Let's look at the rear side, depicted in Figure 2:

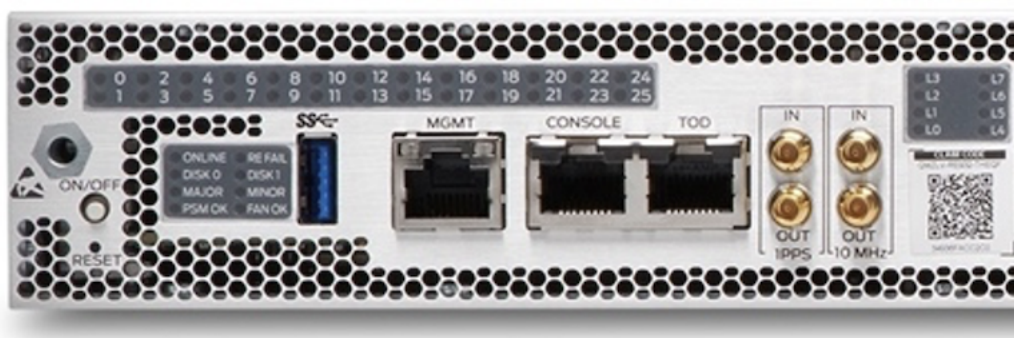
**Figure 2:** MX301 Rear View

The MX301 is equipped with six fans that provide N+1 redundancy, ensuring airflow from front to back. The cooling system is designed to meet NEBS-3 compliance standards.

For power, the chassis includes two power supply units (PSUs) with 1+1 redundancy. The PSUs are highly power-efficient, typically consuming less than 500W, and support AC, DC, or HV input, depending on deployment requirements.

1.4 The Embedded Routing Engine

The MX301 has a built-in routing engine. The RE connectivity is accessible from the front panel (left side). The following figure zooms in on this specific part:

**Figure 3:** MX301 Embedded RE

This built-in RE provides the following resources:

- Intel 10-core (20-thread) CPU, 3.0 GHz, with QAT
- TPM 2.0 module for security (Trusted Platform Module 2.0)

- 128 GB DDR4 memory
- 2x internal 200 GB SSDs
- BITS and GPS clock support
- 1 console port / 1 Ethernet management port
- 1 external USB

An important difference between the MX301 and MX304 Routing Engines is the presence of the Intel QuickAssist Technology (QAT) module. QAT is a hardware accelerator integrated into certain Intel chipsets, CPUs, and add-in cards. Instead of running some algorithms in software on the CPU, QAT uses a dedicated hardware engine to perform them faster and more efficiently, freeing CPU resources for other tasks. QAT accelerates operations such as:

- Cryptography: TLS/SSL, IPsec, disk encryption, VPNs, etc.
- Compression/Decompression: zlib/deflate, gzip, and similar formats.

At the time of writing, the plan is to utilize QAT to enhance the Inline IPsec control plane performance significantly.

Note: This feature leveraging QAT is scheduled for release 26.2, but the timeline may change. Please check with your Juniper or HPE representative for the latest updates.

Another enhancement in the MX301's Routing Engines is the TPM v2 module, which provides stronger security features, improved cryptographic protocols, and secure storage for passwords, encryption keys, and certificates. The main differences between TPM v1 and TPM v2 are summarized in the table below.

Table 3: Main TPM v1.2 and v2 differences

Feature	TPM 1.2	TPM 2.0
Hash algorithms	SHA-1 only	SHA-1, SHA-2 family, SHA-3 (opt)
Public key crypto	RSA only	RSA + ECC
Symmetric crypto	AES-128 only	AES-128/256, SM4, Camellia (opt)
PCRs	24, SHA-1 only	24+, selectable hash algo
Hierarchy model	Single	Multiple (platform, storage, etc.)
NV storage	Limited	Flexible, supports counters/data
Status	Legacy / deprecated	Current & required going forward

The software architecture of this platform differs slightly from that of other Routing Engines. This 1RU system includes a built-in FPC that does not have a dedicated CPU. In comparison, the MX304's built-in FPC and all line cards in the MX10K modular chassis, each have their own CPU.

Instead, the MX301 leverages its enhanced RE's CPU to virtualize the line card microcode, called the Universal Line Card (ULC), in the same way it virtualizes the Junos VM.

The resulting software architecture, implemented directly on the routing engine hardware, is illustrated below.

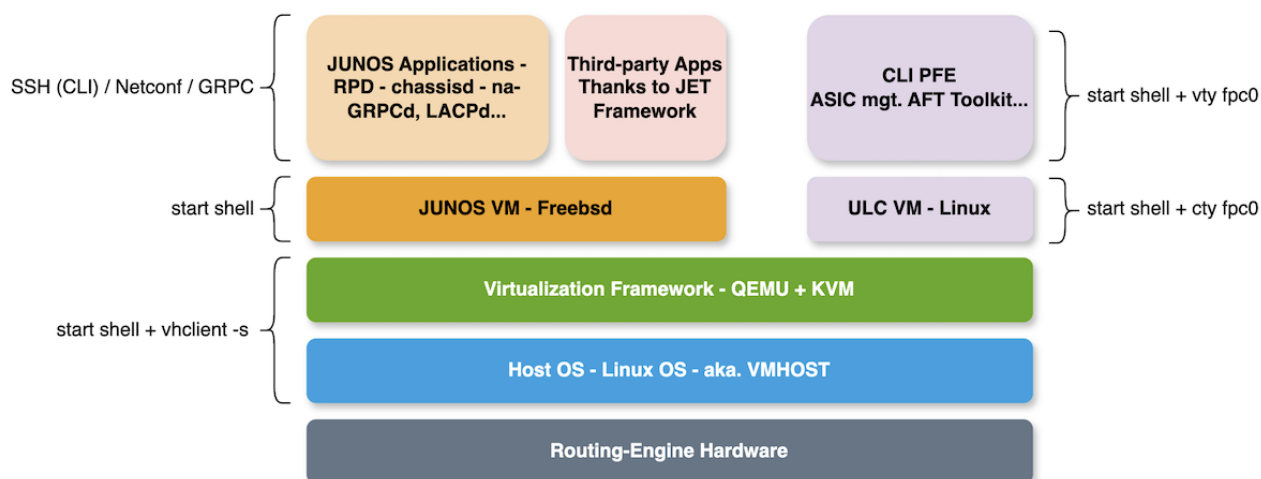


Figure 4: MX301 Software Stack

You can verify this dual-VM architecture by connecting to the host operating system and running the “virsh list” command. You should see two virtual machines:

```
1 bob@mx301> start shell
2 bob@mx301:~ # su
3 bob@mx301:~ # vhclient -s
4 bob@mx301-node:~# virsh list
5 Id Name State
6 -----
7 1 vjunos running
8 2 vULC running
```

As shown, in addition to the standard vJunos VM, there is a second VM named vULC, dedicated to data-plane management. We can use another virsh command to display, for each VM, the resource allocation:

```
1 bob@mx301> start shell
2 bob@mx301:~ # su
3 bob@mx301:~ # vhclient -s
4 Last login: Tue Sep 30 12:38:46 2025 from 192.168.1.2
5 bob@mx301-node:~# virsh dominfo vjunos
6 Id: 1
7 Name: vjunos
8 UUID: 41f2c1f0-fe92-4f3e-a152-8f69a0f39a0a
9 OS Type: hvm
10 State: running
11 CPU(s): 8
12 CPU time: 694182.3s
13 Max memory: 83886080 KiB
14 Used memory: 83886080 KiB
15 <-- truncated output -->
16
17 bob@mx301-node:~# virsh dominfo vULC
18 Id: 2
19 Name: vULC
20 UUID: 433afdfa-5739-43db-8172-c66f3f8f3252
21 OS Type: hvm
22 State: running
23 CPU(s): 6
24 CPU time: 518780.6s
25 Max memory: 33554432 KiB
26 Used memory: 33554432 KiB
27 <-- truncated output -->
```

As shown, the Junos VM has 8 vCPUs and ~80 GB of RAM, whereas the ULC VM has 6 vCPUs and ~32 GB of RAM. The table below summarizes the routing-engine specifications:

Table 4: MX301 RE Specification Summary

Model	MX301-RE
Cores	10
vCPU	20
CPU Clock	3GHz
QAT	Yes
TPM	v2.0
DRAM	128GB
Disks	2x 200GB
Junos VM vCPU	8
Junos VM Mem	~80 GB
ULC VM vCPU	6
ULC VM Mem	~32 GB

1.5 MX301 port connectivity

The MX301 is built around a single Trio 6 ASIC. It offers the same feature set and scaling capabilities as one MX304's LMIC, which is based on the same Trio generation.

Let's verify this:

```

1 bob@mx301> start shell
2 bob@mx301:~ # su
3 bob@mx301:~ # vty fpc0.0
4
5 LNX-FPC0(vty)# show jspec client
6
7 ID Name
8 <-- truncated output -->
9 1 YTCHIP[0]
```

The Trio 6 ASIC is made of two slices or PFE. Each has a capacity of 800Gbps full duplex. We may confirm the presence of the two 800GE PFEs by issuing the following commands:

```

1 bob@mx301> show chassis fpc 0 detail
2 Slot 0 information:
3   State Online
4   Total CPU DRAM 32768 MB
5   Total HBM 8192 MB
6   FIPS Capable True
7   Start time 2025-11-12 07:56:38 PST
8   Uptime 6 minutes, 11 seconds
9   Max power consumption 500 Watts
10  Operating Bandwidth 1600 G
11
12 PFE Information:
13
14   PFE Power ON/OFF Bandwidth SLC
15   0 ON 800G
16   1 ON 800G
```

This single ASIC connects the 26 cages on the front panel:

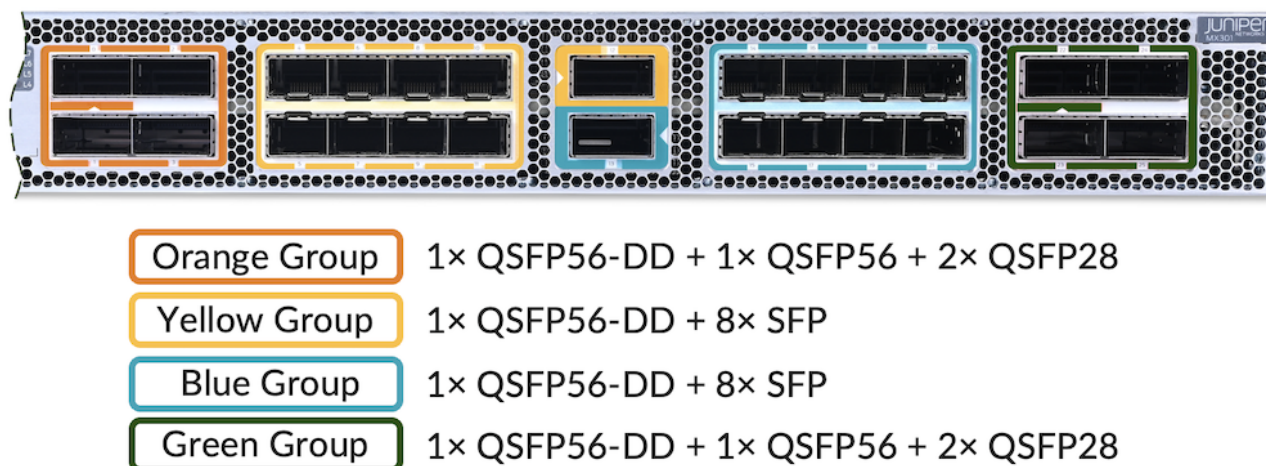


Figure 5: MX301 port groups

As seen in the above picture, there are four groups of ports, also known as port groups, distinguished by color:

- **Orange Group:** 1x QSFP56-DD + 1x QSFP56 + 2x QSFP28
- **Yellow Group:** 1x QSFP56-DD + 8x SFP
- **Blue Group:** 1x QSFP56-DD + 8x SFP
- **Green Group:** 1x QSFP56-DD + 1x QSFP56 + 2x QSFP28

Each port group has a maximum capacity of 400 Gbps and cannot be oversubscribed. This means that, depending on your port speed configuration, some ports may be automatically disabled.

Note: Once available online, please refer to the MX301 port checker for validating the supported configuration.

This is also important to note that each group operates independently. The table below shows the MX301 port density, per interface speed:

Table 5: MX301 Port Density

Interface	Per chassis port density
1GE	32 (breakout)
10GE	32 (breakout)
25GE	32 (breakout)
40GE	6 (Native)
50GE	32 (breakout)
100GE	16 (Native)
400GE	4 (Native)

Note: For native 1G, the MX301 has no restriction and can support jumbo frames up to 12K bytes on 1GE ports.

1.6 Ports Capability

The following Figure 6 provides the direct mapping between each physical port and its port-group and PFE instance:

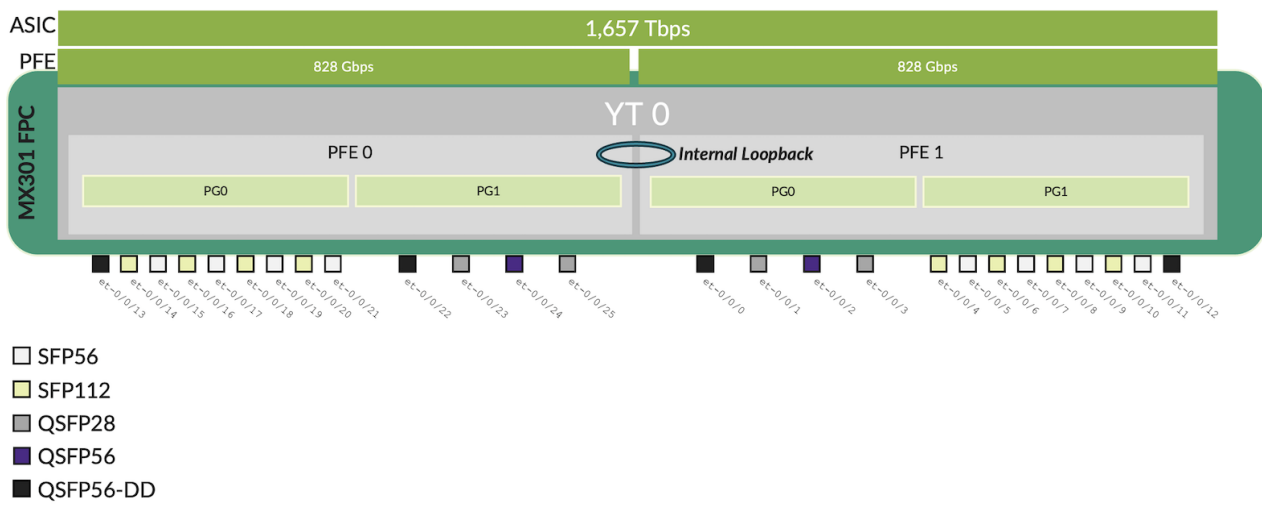


Figure 6: MX301 Internal port mapping

The table below summarizes the PIC port speed capability for the MX301.

Table 6: MX301 Port Speed/Type Mapping

PIC	Ports	Port Type	Port Speed
0	0	QSFP56-DD	1x1GE1x10GE1x25GE8x50GE1x100GE2
0	1, 3	QSFP28	1x1GE1x10GE1x25GE4x1GE4x10GE4x2
0	2	QSFP56	1x1GE1x10GE1x25GE1x100GE
0	4-11	SFP56	1x1GE1x10GE1x25GE1x50GE
0	12	QSFP56-DD	1x1GE1x10GE1x25GE1x40GE8x50GE1x
0	13	QSFP56-DD	1x1GE1x10GE1x25GE1x40GE8x50GE1x
0	14-21	SFP56	1x1GE1x1GE1x25GE1x50GE
0	22	QSFP56-DD	1x1GE1x10GE1x25GE8x50GE1x100GE2
0	23, 25	QSFP28	1x1GE1x10GE1x25GE4x1GE4x10GE4x2
0	24	QSFP56	1x1GE1x10GE1x25GE1x100GE

The CLI below shows similar indications for each port speed and channelization capabilities.

Note: Remember, there is only one PIC (0) since there is a single Trio 6 ASIC.

```

1 bob@mx301> show chassis pic fpc-slot 0 pic-slot 0 <<< FPC and PIC slot are always 0
2 FPC slot 0, PIC slot 0 information:
3   Type MRATE PIC 4x400G/14x100G/8x50G
4   State Ready
5   PIC version 0.0

```

```

6
7 Port speed information:
8
9 Port PFE Port-Group Capable Port Speeds
10 0 1 0 1x1GE, 1x10GE, 1x25GE, 8x50GE, 100GE, 2x100GE, 3x100GE, 4x100GE, 400GE
11 1 1 0 1x1GE, 1x10GE, 1x25GE, 4x1GE, 4x10GE, 4x25GE, 40GE, 100GE
12 2 1 0 1x1GE, 1x10GE, 1x25GE, 100GE
13 3 1 0 1x1GE, 1x10GE, 1x25GE, 4x1GE, 4x10GE, 4x25GE, 40GE, 100GE
14 4 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
15 5 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE
16 6 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
17 7 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE
18 8 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
19 9 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE
20 10 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
21 11 1 1 1x1GE, 1x10GE, 1x25GE, 1x50GE
22 12 1 1 4x1GE, 4x10GE, 4x25GE, 40GE, 8x50GE, 100GE, 2x100GE, 3x100GE, 4x100GE, 400GE
23 13 0 0 4x1GE, 4x10GE, 4x25GE, 40GE, 8x50GE, 100GE, 2x100GE, 3x100GE, 4x100GE, 400GE
24 14 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
25 15 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE
26 16 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
27 17 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE
28 18 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
29 19 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE
30 20 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE, 100GE
31 21 0 0 1x1GE, 1x10GE, 1x25GE, 1x50GE
32 22 0 1 1x1GE, 1x10GE, 1x25GE, 8x50GE, 100GE, 2x100GE, 3x100GE, 4x100GE, 400GE
33 23 0 1 1x1GE, 1x10GE, 1x25GE, 4x1GE, 4x10GE, 4x25GE, 40GE, 100GE
34 24 0 1 1x1GE, 1x10GE, 1x25GE, 100GE
35 25 0 1 1x1GE, 1x10GE, 1x25GE, 4x1GE, 4x10GE, 4x25GE, 40GE, 100GE

```

The default port speed for all ports is 100 Gbps, with interface names beginning with “et-”. As on the MX304 LMIC16, you can enable different speeds and/or channelization using the standard set of commands:

- **set chassis fpc 0 pic 0 pic-mode** - configure speed at the chassis level
- **set chassis fpc 0 pic 0 number-of-sub-ports [0-7]** - configure channelization at the chassis level
- **set chassis fpc 0 pic 0 port [port-num] speed** - configure speed at the port level
- **set chassis fpc 0 pic 0 port [port-num] number-of-sub-ports [0-7]** - configure channelization at the port level

Note: FPC slot and PIC slot are always 0 on the MX301 chassis.

On MX, the port name prefix or suffix depends on the configured speed and whether channelization is enabled. The table below reminds you the naming conventions for different speeds:

Table 7: MX301 Port Naming Based on Speed

Speed	Port Name
1GE	ge-x/y/z
4x1GE	ge-x/y/z:0-3
10GE	xe-x/y/z
4x10GE	xe-x/y/z:0-3
25GE	et-x/y/z
4x25GE	et-x/y/z:0-3
40GE	et-x/y/z
50GE	et-x/y/z

Speed	Port Name
8x50GE	et-x/y/z:0-7
100GE	et-x/y/z
2x100GE	et-x/y/z:0-1
3x100GE	et-x/y/z:0-2
4x100GE	et-x/y/z:0-3
400GE	et-x/y/z

1.7 Example of port configuration

For this section, we refer to this picture where the four port-groups are highlighted and numbered from 1 to 4:

- PG 1 and PG 4 are made of 4 physical ports
- PG 2 and PG 3 are made of 9 physical ports

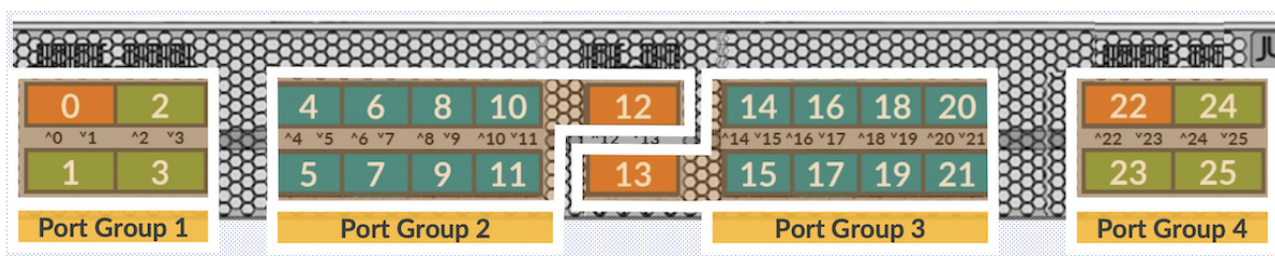


Figure 7: MX301 port speed combinations

Let's first focus on PG 1 and PG 4. The figure below shows six examples of port combinations. As you can see, depending on the configuration, some ports will be automatically disabled to avoid oversubscription. As mentioned earlier, this mechanism is managed per port group.

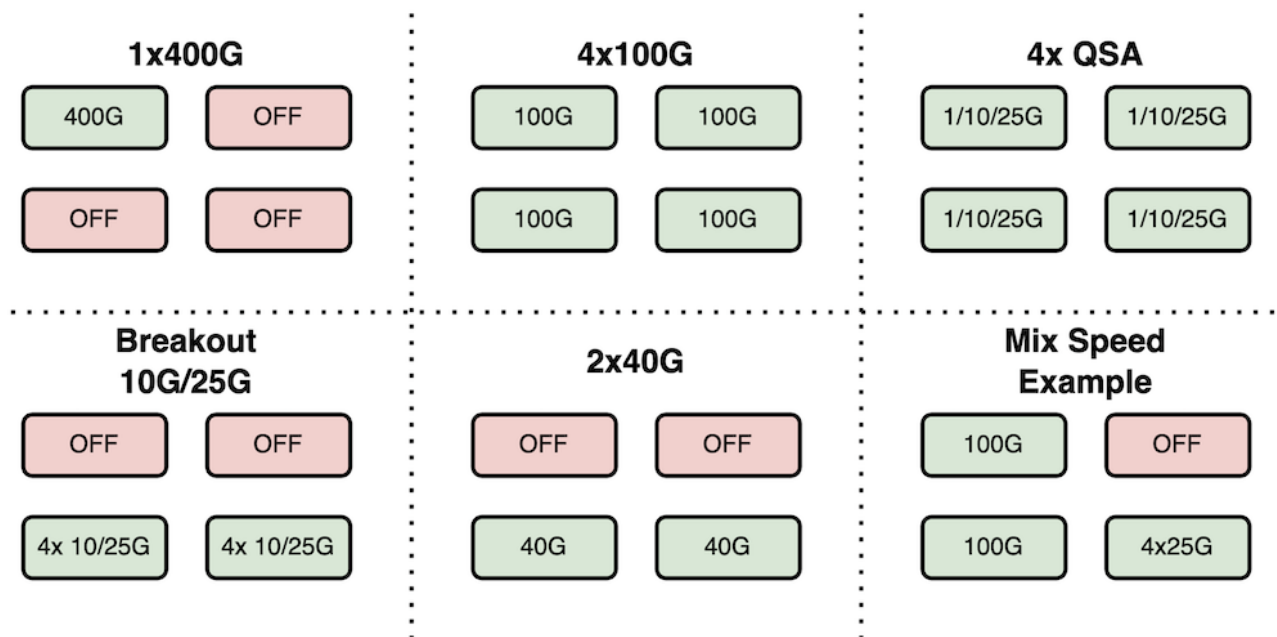


Figure 8: PG 1 or PG 4 sample speed combinations

The following figure provides similar examples but this time for port group 2 or 3:

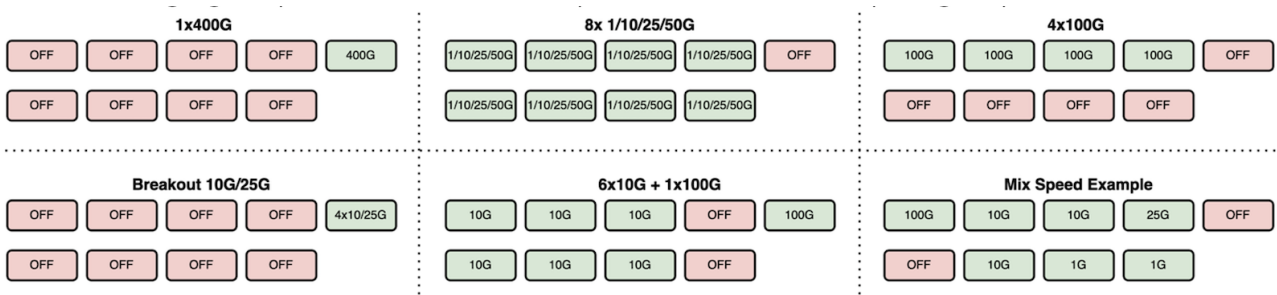


Figure 9: PG 2 or PG 3 sample speed combinations

1.8 Maximum Port Density

1.8.1 400GE

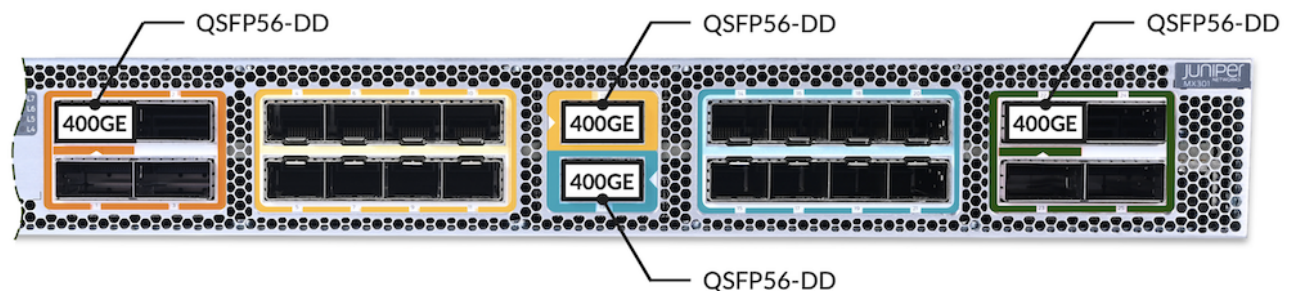


Figure 10: Max number of 400GbE

- 4x 400GE QSFP56-DD

1.8.2 100GE

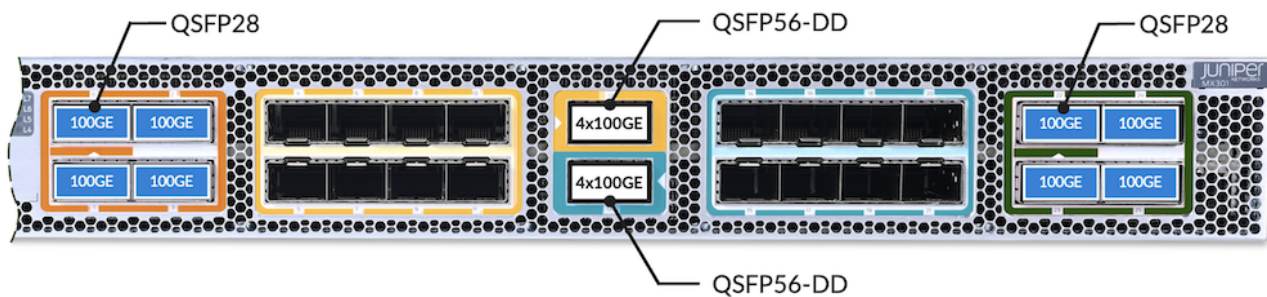


Figure 11: Max number of 100GbE

- 16x 100GE 8x QSFP28 (Native) + 2x QSFP56-DD (BreakOut 4x100GE)

1.8.3 50GE

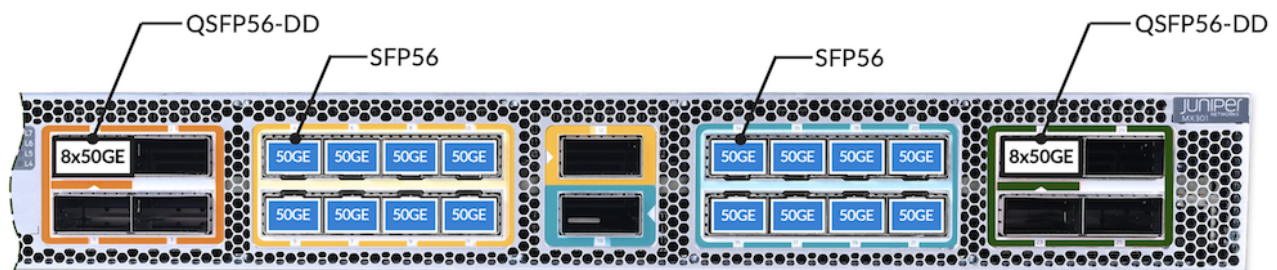


Figure 12: Max number of 50GbE

- 32x 50GE 16x SFP56 (Native) + 2x QSFP56-DD (BreakOut 8x50GE)

1.8.4 40GE

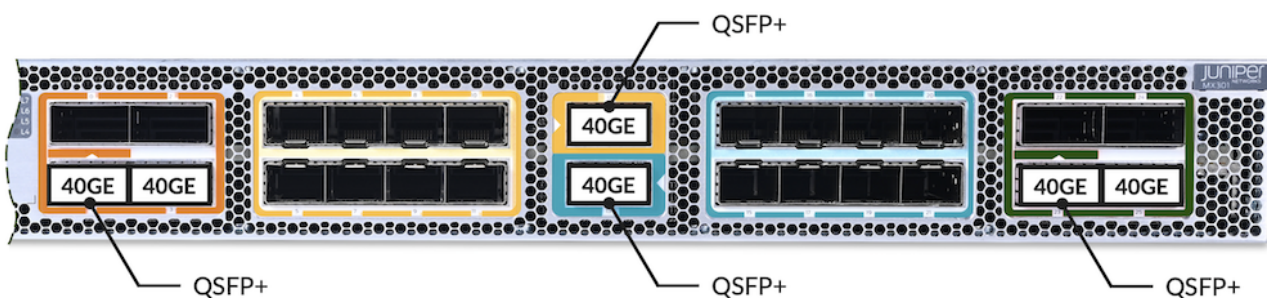


Figure 13: Max number of 40GbE

- 6x 40GE QSFP56-DD

1.8.5 25GE

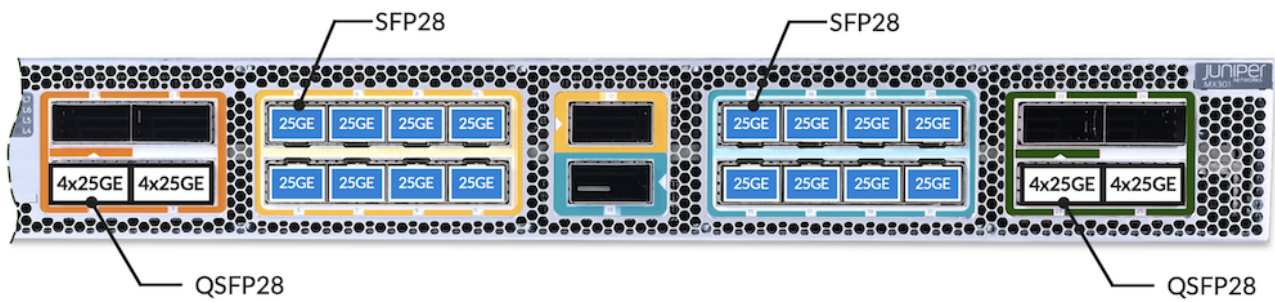


Figure 14: Max number of 25GbE

- 32x 25GE 16x SFP28 (Native) + 4x QSFP28 (BreakOut 4x25GE)

1.8.6 10GE

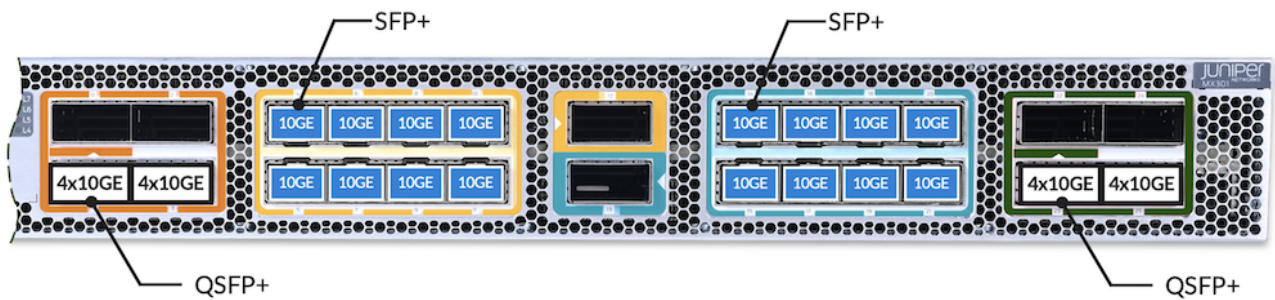


Figure 15: Max number of 10GbE

- 32x 10GE 16x SFP+ (Native) + 4x QSFP+ (BreakOut 4x10GE)

1.8.7 1GE

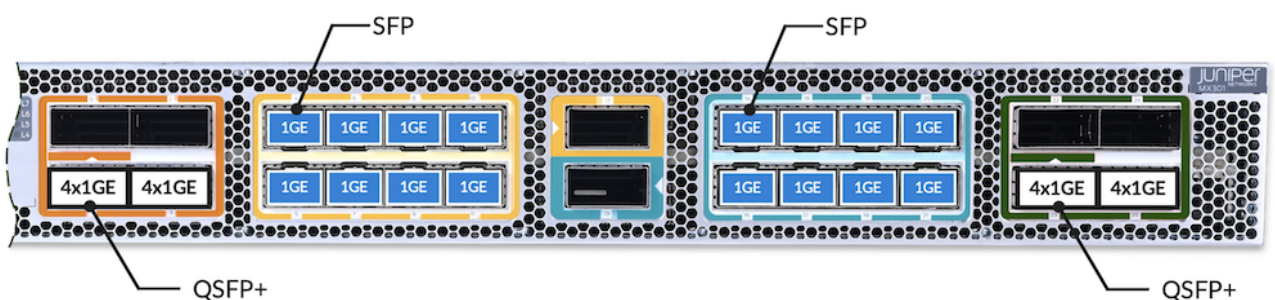


Figure 16: Max number of 1GbE

- 32x 1GE 16x SFP (Native) + 4x QSFP+ (BreakOut 4x1GE)

1.9 MX301 Internal Architecture

In this section, we will examine the internal architecture of the MX301, as shown in the figure below.

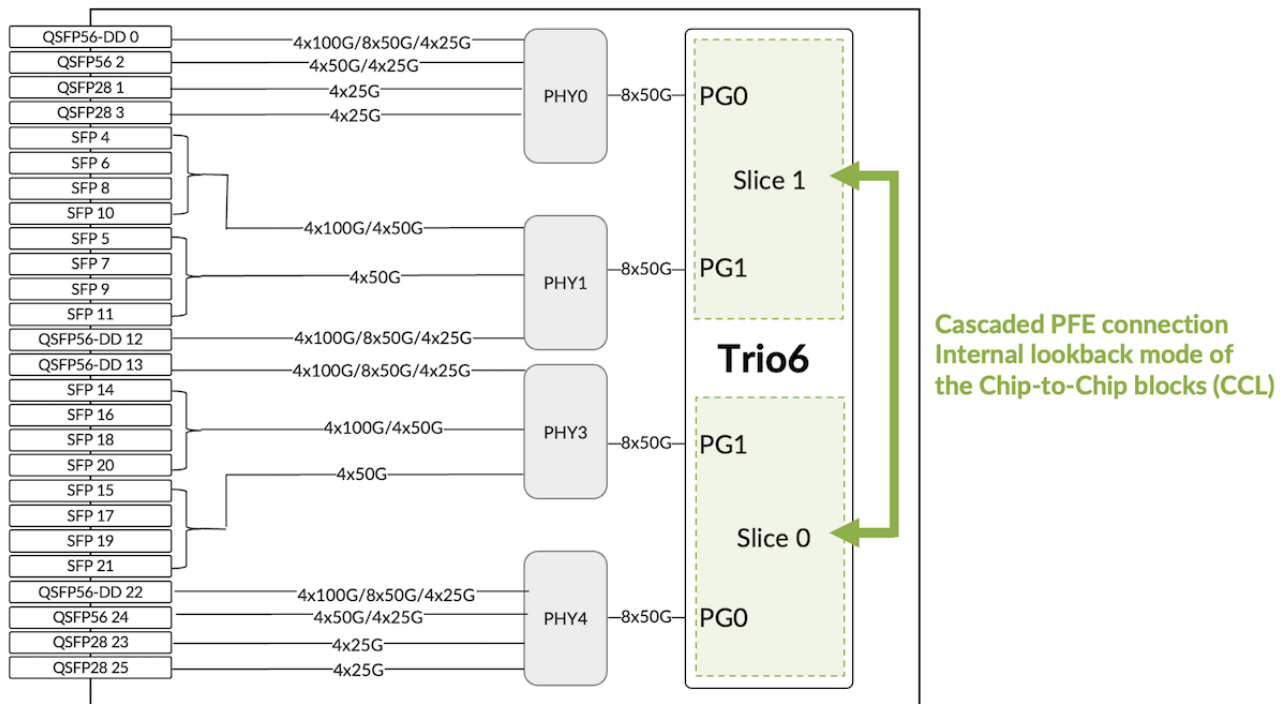


Figure 17: MX301 internal view

As you can see, there is a single Trio 6 chipset that integrates the two PFEs in one package. The sixth-generation Trio chipset (also known as YT) operates in Run-to-Completion mode. This ASIC features several innovations, including a new filtering acceleration block and an embedded crypto engine for inline IPsec. The MX301 also supports Class C timing and provides line-rate MACsec on all ports. The chassis can handle up to 10 million prefixes in the FIB, H-QoS with up to five levels, and 128K queues

Note: For more information on the Trio 6 ASIC, see Techpost [1]

In Figure 10, there are also PHY components. The PHYs are additional components, part of the data plane, and interconnected to the PFEs. PHY works closely with SerDes and is responsible for adapting, boosting, or converting signals between the PFE and the network interfaces. A SerDes (Serializer/Deserializer) is a high-speed circuit that converts parallel data into serial data for transmission over a single high-speed lane.

PHYs are generally classified into three types:

- **GearBoxes** (or Forward GearBoxes): adapt the SerDes speed and encoding to connect to higher-speed ports.
- **Reverse GearBoxes** (RGBs): adapt the SerDes speed and encoding to connect to lower-speed links.
- **Retimers**: do not change the SerDes speed or encoding but amplify the signal to extend its reach.

In the MX301, we use PHY as GearBoxes and/or Reverse GearBoxes to offer a wide range of port speeds.

As also shown in Figure 10, the MX301 does not include a separate fabric chip. This is a Fabric-Less design.

1.10 Conclusion

The MX301 expands the MX10K Series by providing a compact 1RU solution while retaining the power and features of larger models, such as the MX304 or the LC9600 MX10K. It is built on Trio 6 technology, delivering

a complete set of routing and switching capabilities. With a full-duplex throughput of 1.6 Tbps, the MX301 can support a wide range of demanding use cases, including enterprise networks, regional peering routers, Route Reflectors, and aggregation scenarios, as well as BNG.

In summary, the MX301 combines performance, compactness, and versatility, extending the capabilities of the MX10K Series while adapting to diverse deployment requirements.

1.11 Useful links

- [1] <https://juniper.github.io/techpost/articles/trio-6-packet-walkthrough/article>

1.12 Glossary

- ASIC: Application-Specific Integrated Circuit
- BITS: Building Integrated Timing Supply
- BNG: Broadband Network Gateway
- CPE: Customer Premises Equipment
- Class C Timing: ITU-T class timing compliance level
- CLI: Command-Line Interface
- DC: Data Center
- DC-GW: Data Center Gateway
- FIB: Forwarding Information Base
- FPC: Flexible PIC Concentrator (built-in line card module)
- FRU: Field-Replaceable Unit
- GPS: Global Positioning System (timing/clock source)
- H-QoS: Hierarchical Quality of Service
- IPsec: Internet Protocol Security
- LC9600: MX10K line card model reference
- LMIC: Line-Module Integrated Card
- MACsec: Media Access Control Security (layer-2 encryption)
- MX10K: Juniper MX10000 Series
- MX301: Juniper Networks MX301 Universal Routing Platform
- NEBS-3: Network Equipment-Building System Level 3 compliance
- PE: Provider Edge
- PFE: Packet Forwarding Engine
- PHY: Physical Layer device
- PIC: Physical Interface Card
- PSM: Power Supply Module
- PSU: Power Supply Unit
- QAT: Intel QuickAssist Technology (crypto/compression acceleration)
- QSFP: Quad Small Form-factor Pluggable
- RE: Routing Engine
- SerDes: Serializer/Deserializer
- SFP: Small Form-factor Pluggable
- Tbps: Terabits per second

- TLS/SSL: Transport Layer Security / Secure Sockets Layer
- TPM: Trusted Platform Module
- ULC: Universal Line Card (virtualized data-plane microcode VM)
- vCPU: Virtual CPU
- VM: Virtual Machine
- VPN: Virtual Private Network
- WAN: Wide Area Network

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